7.9 Using Trigonometry in the Real World

Lesson Introduction

Benchmark	MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem	Learning Target	• I can solve real-world problems using trigonometric ratios and the Pythagorean Theorem.

	Building On:	Working Toward:
Benchmark Coherence	MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.	N/A

Essential Questions	• How can I solve real-world problems using trigonometric ratios and the Pythagorean Theorem?
Lesson Narrative	In this lesson, students are introduced to trigonometry in the real world in scenarios involving the angle of depression and the angle of elevation. Students begin by exploring the general concept of the angle of depression and the angle of elevation, prior to learning how to determine the value of those angles to solve real-world problems.

Component Summary	7.9.1 Warm-Up (~5 min.)	Notice and Wonder about a spiral staircase *(SE p. 422)*	7.9.4 Guided Practice (10-15 min.)	Solving using trigonometric ratios with angle of elevation and angle of depression *(SE p. 425)*
	7.9.2 Collaboration (5 min.)	Exploring angle of depression and angle of elevation *(SE p. 422)*	7.9.5 Your Turn! (5-10 min.)	Independent practice solving real-world problems using trig *(SE p. 426)*
	7.9.3 Guided Instruction (5-10 min.)	Solving with angle of elevation and angle of depression *(SE p. 423)*	7.9.6 Wrap-Up (~5 min.)	Reflecting on gaps from the lesson and strategies to fill them *(SE p. 427)*

	Glossary	Materials	Additional Preparation
Resources	Key Terms: • Angle of Elevation • Angle of Depression Additional Terms: N/A	• Ruler • (optional) String or Yarn	• This lesson consists almost entirely of real-world word problems. Consider determining which word problem strategy would best be modeled and implemented with your students.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may switch the values of the angle of elevation and the angle of depression.
Students may use the value of the wrong angle as the value of the angle of elevation or the angle of depression.</td><td>The structure of the lesson is to have students collaborate to understand the general concept of the angle of depression and the angle of elevation in 7.9.2 Collaboration. Students will learn to find these values in 7.9.3 Guided Instruction. They will apply that, along with previously learned methods, to solve real-world problems for the remainder of the unit.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Notice and Wonder
The warm-up provides students with an image of a spiral staircase. They are asked to notate what they notice from the image and what they wonder from the image, along with a more specific wondering about capturing the number of stairs. This routine provides a low entry point for students to engage in general discussion.</td><td>MA.K12.MTR.7.1: Real-World Contexts
From 7.9.3 Guided Instruction and beyond, students will focus on applying trigonometry to solve real-world problems. Through this, students will connect mathematical concepts to everyday experiences and use models and methods to understand, represent, and solve problems.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Physical manipulatives and movement can help students understand angle of elevation and angle of depression. Consider having a student stand on a platform and hold a piece of yarn. Have another student stand on the ground and hold the other end of the yarn. Use this to show where the angle of elevation is and the angle of depression is, so that students have a more engaging visual. This will also help kinesthetic learners by having them interact with angle of elevation and angle of depression.

Additional differentiation opportunities are denoted throughout the lesson guide.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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7.9.1 Warm-Up: Notice and Wonder

Component Overview: *Students will describe what they notice and what they wonder about a provided image.*

7.9.1 Warm-Up: Notice and Wonder

1. The picture shows the stairs in the Ponce de Leon Inlet Lighthouse.

   a. What do you notice?

   Answers will vary. It looks like a shell.

   b. What do you wonder?

   Answers will vary. I wonder how many stories there are.

   c. The lighthouse has 203 steps. Do you think this picture includes all of the steps? Explain your answer.

   Answers will vary. Yes, I can see all of the steps if I look straight down the center.

This activity provides students with an image of a spiral staircase. They are asked to notate what they notice from the image and what they wonder from the image, along with a more specific wondering about capturing the number of stairs. This routine provides a low entry point for students to engage in general discussion.

7.9.2 Collaboration: Angle of Depression and Angle of Elevation

Component Overview: *Students will collaborate to discover the concept of an angle of depression and an angle of elevation. For this activity, students will need a partner or small group.*

7.9.2 Collaboration: Angle of Depression and Angle of Elevation

Real-world trigonometric ratio problems involve angles that are made from a horizontal line of sight.

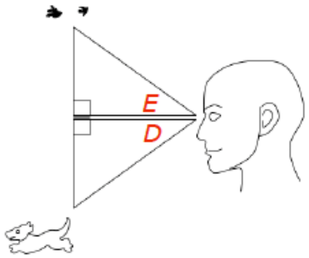
1. Using a ruler, draw the horizontal line of sight for the man.

For this activity, students will need a ruler in order to complete the step in question 1.

2. The horizontal line of sight is always [○ perpendicular / ○ parallel] to the ground.

3. The angles created from the horizontal line of sight are called angles of elevation and angles of depression. In the image, Hector is looking at birds in the sky and a dog on the ground.

   Label the angle of elevation with an E, label the angle of depression with a D.



How did you choose which one was the angle of elevation, and the other was the angle of depression?

7.9.2 Collaboration: Angle of Depression and Angle of Elevation (continued)

4. Based on the name of the angle, work with a partner to write a definition for an angle of elevation.
   The angle of elevation is the angle created above the horizontal when looking up at an object.

5. Work with a partner to write a definition for an angle of depression.
   The angle of depression is the angle created below the horizontal when looking down at an object.

6. Compare your definitions with that of another pair. Discuss any differences. Make changes as necessary.
   Answers will vary.

At the conclusion of this activity, consider having students share out their definitions of the angle of elevation and the angle of depression. Record student responses on chart paper or a white board to use as a reference point when they are introduced to the formal definitions and concept in the next activity.

7.9.3 Guided Instruction: Angles of Elevation and Depression

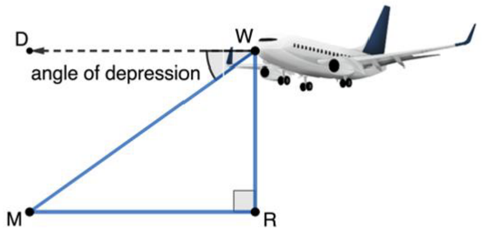
Component Overview: *Students will be introduced to the formal definitions of the angle of elevation and the angle of depression and how to use those values to solve a real-world problem.*

7.9.3 Guided Instruction: Angles of Elevation and Depression

The **angle of elevation** is the angle created above the horizontal when looking up at an object.

The **angle of depression** is the angle created below the horizontal when looking down at an object.

1. The angle of depression, $m\angle DWM$, from the airplane to the runway, point M, is $37°$.



a. Explain why $\angle DWM$ is congruent to $\angle RMW$.
Because $\overline{MR}$ is parallel to $\overline{DW}$ and Alternate Interior Angles are congruent.

b. If the airplane is flying at an altitude of 2500 feet, WR, and is descending to the airport, M, what is the distance, MW, the plane has to travel?

$sin(37)° = \frac{2500}{MW}$; $MW \approx 4154.1$ *feet*

c. How far is the runway from a point on the ground directly below the airplane, MR?
$(2500)^2 + (MR)^2 = (4154.1)^2$
$MR = 3317.612$ *feet*

Physical manipulatives and movement can help students understand angle of elevation. Consider having a student stand on a platform and hold a piece of yarn, while another student stands on the ground ands hold the other end of the yarn. Use this to show where the angle of elevation and the angle of depression are so that students have a more engaging visual.

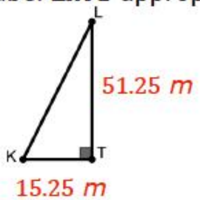
Consider modeling for students the labeling of the triangle with the information about the altitude value in question 1b, but releasing them to try and figure out the answer with a partner. Remind them that they have all the information they need to find the correct value. Bring the students back together to review before releasing students to try question 1c on their own.

7.9.3 Guided Instruction: Angles of Elevation and Depression (continued)

2. Kamala is visiting the Ponce de Leon Inlet Lighthouse. Kamala is standing at point K and is looking at the top of the lighthouse, point L. $\overline{KT}$ is from Kamala's eye level, which is at 1.75 meters (m).

   a. When Kamala looks at the top of the lighthouse, is the angle created an angle of elevation or depression?
   Angle of elevation.

   b. The height of the Ponce de Leon Inlet Lighthouse is 53 m. Kamala is standing 15.25 m from the lighthouse. Label ΔKTL appropriately with this information.



   c. What is the measure of the angle created when Kamala looks at the top of the lighthouse?
   $m\angle K = tan^{-1}\left(\frac{51.25}{15.25}\right)^{\circ} \approx 73.429^{\circ}$

- *Why is this an example of an angle of elevation, and not an angle of depression?*
- *Where would Kamala need to be standing to create an angle of depression?*

What would be the first step in solving this problem? Why is that first step so important?

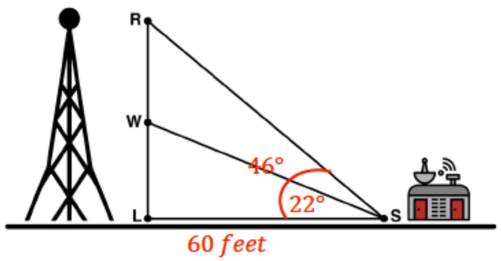
Students may need to be reminded to label their triangle with the given information, so they can see that they have the information needed to solve.

7.9.4 Guided Practice: Trigonometry in the Real World

Component Overview: *Students will complete two real-world problems where they will apply what they have learned about trigonometric ratios.*

7.9.4 Guided Practice: Trigonometry in the Real World

1. A cell phone tower was built in two sections, with a control shed built 60 feet from the base of the tower on level ground. From the base of the control shed, the angle of elevation, $\angle LSW$, to the top of the first section of tower, $\overline{LW}$, is $22°$. The angle of elevation to the top of the second section, $m\angle LSR$, is $46°$.



   a. Label the diagram with the information from the problem.
   See above.

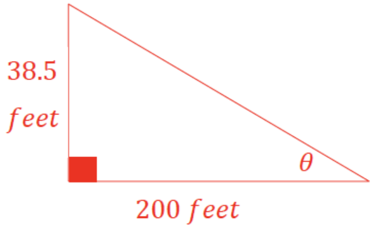
   b. Determine the length of $\overline{RW}$, the second section of tower. Round your answer to the nearest thousandth of a foot.
   $tan(22°) = \frac{WL}{60}$; $WL \approx 24.242$ feet
   $tan(46°) = \frac{RL}{60}$; $RL \approx 62.132$ feet
   $RW = RL - WL \approx 37.89$ feet

For this activity, consider breaking it up into two problems. Have students complete question 1 independently and then bring them back together. Have students come to the board (or other method) and label one thing in the diagram. Repeat this with multiple students until all labeling has been completed. Then have different students come up to show a step in determining the length. Repeat until all steps have been shown. Engage students in a discussion to better understand any potential misconceptions before moving onto question 2.

7.9.4 Guided Practice: Trigonometry in the Real World (continued)

2. A hot air balloon is flying 38.5 ft above level ground. The Orlando City Soccer field is 200 ft from the point on the ground directly below the balloon.

   a. Draw and label a diagram using the information from the problem.



   b. What is the angle of elevation from the soccer field to the hot air balloon?

   $\theta = tan^{-1}\left(\frac{38.5}{200}\right)^{\circ} \approx 10.896^{\circ}$

   c. The hot air balloon ascends further into the air. If the new angle of elevation from the soccer field to the balloon is $17.3°$. How many feet did the hot air balloon rise?

   $tan(17.3)^{\circ} = \frac{height}{200}$; $height \approx 62.293$ feet
   $62.293 - 38.5 = 23.793$
   The balloon rose an additional 23.793 *feet.*

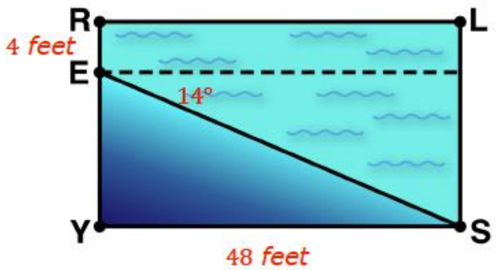
For question 2, consider leveraging a Think-Pair-Share instructional routine. Give students a set amount of independent processing time to work through the problem on their own to see what they can do independently. Then call "time," and have students work together as partners to calibrate their responses and come to a consensus on the correct answers, prior to reviewing as a whole class.

7.9.5 Your Turn!

Component Overview: *Students work independently to practice comparing while using trigonometric ratios to solve a real-world problem.*

7.9.5 Your Turn!

1. A swimming pool is 48 feet long and 4 feet deep in the shallow end. There is a steady decline of $14°$ toward the deep end of the pool. Rectangle $RLSY$ and ΔSYE are shown to model the transition from the shallow end to the deep end. Label the diagram appropriately.



   a. How deep is the pool at the deep end?

   $tan(14)° = \frac{EY}{48}$, $EY \approx 11.968$ feet

   $LS = RE + EY \approx 15.968$ feet

   b. $\overline{ES}$ represents the slant in the pool from the shallow end to the deep end. What is its length?

   $cos(14)° = \frac{48}{ES}$; $ES \approx 49.469$ feet

Consider using this time to pull together a small group of students who may have been struggling during the guided instruction or guided practice portions of this lesson to provide additional support with this problem in a small group setting.

For students who finish early, challenge them to create a word problem that could be solved by using trigonometric ratios. Have them trade their problem with a partner to see if their partner can solve it.

7.9.6 Wrap-Up: Reflection

***Component Overview**: Students complete a self-assessment based on their level of understanding of using an angle of depression or an angle of elevation.*

7.9.6 Wrap-Up: Reflection

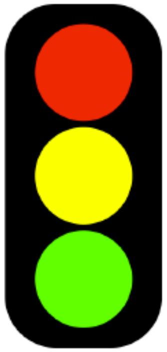
1. Circle the color on the stoplight that best represents your understanding of solving real-world problems using an angle of depression or an angle of elevation.

   Answers will vary.

   Red: I do not understand.

   Yellow: I need some more help.

   Green: I got this; I understand!



Consider using student responses from prior self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.

2. Complete the following statements if you selected red or yellow.

   a. I am still unsure about ...

   b. To fill in this gap I intend to ...